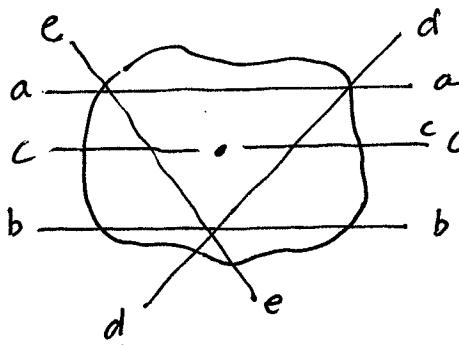


Parallel Axis Theorem (PAT): Derivation

(from Hibbeler Ch. 10.2)



At left is a general area, A.

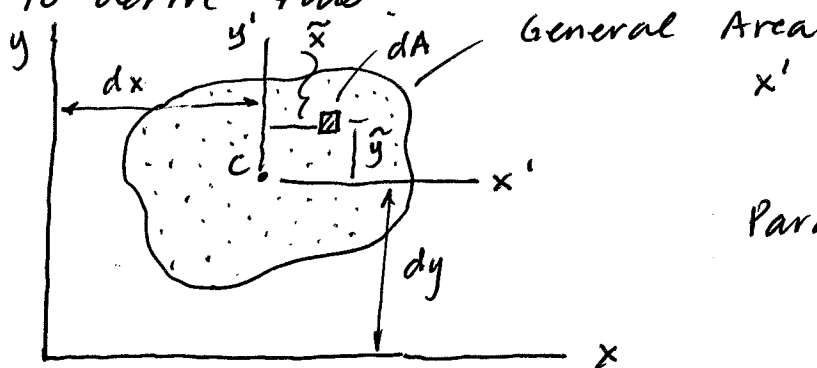
It has a different (Area) moment of inertia

about every axis: $\begin{cases} a-a \\ b-b \\ c-c \\ d-d \\ e-e \end{cases}$

About axes parallel to others, and about rotated axes.

In order to determine the MOI of an area about an axis that is parallel to the centroidal axis, we use the Parallel Axis Thm. (I call it "PAT")

To derive this:



x' and y' are centroidal axes

Parallel to x and y.

$$I_x = \int_A (d_y + \tilde{y})^2 dA$$

This is $\bar{I}_{x'}$

$$= \int_A \tilde{y}^2 dA + 2d_y \int_A \tilde{y} dA + d_y^2 \int_A dA$$

Since $\tilde{y}$ are measured from centroidal axis

Thus, $\boxed{\bar{I}_x = \bar{I}_{x'} + A d_y^2}$ — Book

I write it this way.

or $\boxed{I_x = \bar{I} + A d_y^2}$ $\bar{I}$ implies centroid, so the x' subscript is redundant.

Think about the PAT:

$$I_x = \bar{I} + A d_y^2$$

Implied that $\bar{I}$ about centroidal axis parallel to x.

or

$$I_y = \bar{I} + A d_x^2$$

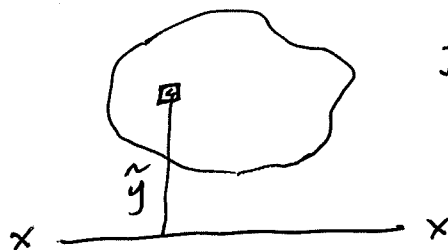
$\bar{I}$ parallel

- ① The term $A d^2$ is always positive, so $\bar{I}$ is a minimum ... about the centroidal axis.

- ② Since $I_x = \int_A y^2 dA$, if the axis passes thru the centroid, the y values (and thus y^2) are minimized.



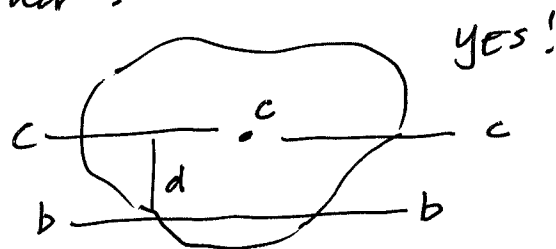
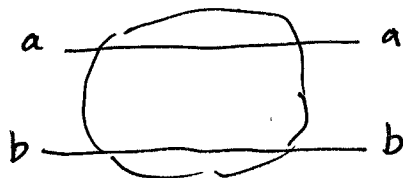
$$I_{cc} = \bar{I} = \text{Minimum}$$



$$I_{xx} > \bar{I} = I_{cc}$$

- ③ The PAT cannot be used from one general parallel axis to another:

NO!



One of the axes must be the centroidal axis.

* CANNOT write

$$I_{aa} = I_{bb} + A d^2$$

* CAN WRITE:

$$I_{bb} = \bar{I} + A d^2$$

"I_{cc}"